Assignment 4

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Please note that

- **Released date: 13th March, Wednesday.**
- **Due date: 23th March, Saturday, by 11:55 pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS1.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. Determine whether the following is a basis of $\mathbb{R}^3$.

$$\mathcal{U} = \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Solution

i. We know that $\dim(\mathbb{R}^3) = 3$, so

$$\mathcal{U} \text{ is a basis of } \mathbb{R}^3 \Leftrightarrow \mathcal{U} \text{ is linearly independent}$$

ii. Let $c_1, c_2, c_3 \in \mathbb{R}$, such that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = 0$$

iii. $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = 0$ means

$$\begin{cases} c_1 - c_2 + c_3 = 0 \\ \qquad\quad c_3 = 0 \\ c_1 + c_2 \quad = 0 \end{cases} \Rightarrow \begin{cases} c_3 = 0 \\ c_2 = c_1 \\ c_2 = -c_1 \end{cases} \Rightarrow c_1 = c_2 = c_3 = 0$$

Then, $\mathcal{U}$ is basis of $\mathbb{R}^3$.

Question 2. Let $\mathcal{V}$ be a vector space, and consider a linearly independent set $\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2\} \in \mathcal{V}$.

(a) Determine whether $\{\mathbf{u}_1, \mathbf{u}_2, -\mathbf{u}_1 + 2\mathbf{u}_2\}$ is a basis of $\mathcal{V}$

(b) Let $\mathcal{W} = \text{Span}(\{\mathbf{u}_1, \mathbf{u}_2, -\mathbf{u}_1 + 2\mathbf{u}_2\})$. Show that $\dim \mathcal{W} = 2$.

Solution

(a) No. Since $\{\mathbf{u}_1, \mathbf{u}_2, -\mathbf{u}_1 + 2\mathbf{u}_2\}$ is not linearly independent. In fact,

$$1\mathbf{u}_1 + (-2)\mathbf{u}_2 + 1(-\mathbf{u}_1 + 2\mathbf{u}_2) = 0.$$

(b) $\mathbf{U}$ is a basis of $\mathcal{W}$ because

i. $\mathbf{U}$ is linearly independent

ii. If $\mathbf{x} \in \mathcal{W}$, then $\mathbf{x} = k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + k_3(-\mathbf{u}_1 + 2\mathbf{u}_2) = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$, where $c_1 = k_1 - k_3$, $c_2 = k_2 + 2k_3$

Question 3. Let $\mathcal{W}$ be the solution set of the following system,

$$\begin{bmatrix} 1 & -2 & 1 & 2 \\ -1 & 6 & 1 & 3 \\ 3 & 4 & -2 & 1 \end{bmatrix} \mathbf{x} = \mathbf{0}$$

Find a basis of $\mathcal{W}$.

Solution

i. Let $\mathbf{A} = \begin{bmatrix} 1 & -2 & 1 & 2 \\ -1 & 6 & 1 & 3 \\ 3 & 4 & -2 & 1 \end{bmatrix}$ denote the 3×4 matrix, then $\mathcal{W} = \text{N}(\mathbf{A}) = \text{Null}(\mathbf{A})$ is the null space of $\mathbf{A}$.

ii. Transform $\mathbf{A}$ to the reduced row echelon form (REF):

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3/8 \\ 0 & 0 & 1 & 7/4 \end{bmatrix}.$$

So $\left\{ \begin{bmatrix} -1 \\ -3/8 \\ -7/4 \\ 1 \end{bmatrix} \right\}$ is a basis of $\mathcal{W}$ and $\dim \mathcal{W} = 1$.

Question 4. (a) Show that $\mathcal{B} = \{x, x-1, x^2+1\}$ is basis of P_2.

(b) Write the transition matrix from $\mathcal{B}$ to the standard basis of P_2.

(c) Write the transition matrix from the standard basis of P_2 to the basis of $\mathcal{B}$.

(d) Find the coordinates of $p = 1 + 2x + x^2$ in $\mathcal{B}$.

Solution

(a) We know that $\dim P_2 = 3$. It suffice to prove that $\mathcal{B} = \{q_1 = x, q_2 = x-1, q_3 = x^2+1\}$ is linearly independent.

Let $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$c_1 q_1 + c_2 q_2 + c_3 q_3 = 0.$$

So $-c_2 + c_3 + (c_1 + c_2)x + c_3 x^2 = 0 \Rightarrow$

$$\begin{cases} -c_2 + c_3 = 0 \\ c_1 + c_2 = 0 \\ c_3 = 0 \end{cases} \Rightarrow c_1 = c_2 = c_3 = 0$$

Then $\mathcal{B}$ is linearly independent.

(b) The standard basis of P_2 is $\mathcal{U} = \{p_1 = 1, p_2 = x, p_3 = x^2\}$. The transition matrix from $\mathcal{B}$ to $\mathcal{U}$ is

$$\mathbf{A} = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3],$$

where

$$\mathbf{u}_1 = [q_1]_{\mathcal{U}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_2 = [q_2]_{\mathcal{U}} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_3 = [q_3]_{\mathcal{U}} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Then we have

$$\mathbf{A} = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(c) The transition matrix from $\mathcal{U}$ to $\mathcal{B}$ is

$$\mathbf{M} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3],$$

where $\mathbf{v}_1 = [q_1]_B, \mathbf{v}_2 = [q_2]_B, \mathbf{v}_3 = [q_3]_B$.

We have

$$p_1 = 1 = 1(x) + (-1)(x-1) + 0(x^2+1) = 1q_1 + (-1)q_2 + 0q_3$$
$$p_2 = x = 1(x) + 0(x-1) + 0(x^2+1) = 1q_1 + 0q_2 + 0q_3$$
$$p_3 = x^2 = (-1)(x) + 1(x-1) + 1(x^2+1) = (-1)q_1 + 1q_2 + 1q_3$$

So

$$\mathbf{M} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Alternatively, we can use $\mathbf{M} = \mathbf{A}^{-1}$ and calculate the inverse of $\mathbf{A}$.

(d) Let $p = 1 + 2x + x^2$. Then

$$[p]_{\mathcal{U}} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix},$$

and

$$[p]_{\mathcal{B}} = \mathbf{M}[p]_{\mathcal{U}} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Alternatively, we can guess that

$$p = 1 + 2x + x^2 = 2x + x^2 + 1 = 2q_1 + 0q_2 + 1q_3.$$

But for complicated problem this is not recommended.

Question 5. Let $\mathcal{W}$ be the subspace of P_3 spanned by $\{-x, x+1, x^2-1, x^2+1\}$

(a) Does $x^3 \in \mathcal{W}$?

(b) Find a basis of $\mathcal{W}$.

(c) What is $\dim \mathcal{W}$?

Solution

(a) No. Since $p = x^3$ has degree 3, while any linear combination of $q_1 = -x, q_2 = x + 1, q_3 = x^2 - 1$ and $q_4 = x^2 + 1$ would have degree less than two.

(b) i. Let $\mathcal{U} = \{p_1 = 1, p_2 = x, p_3 = x^2, p_4 = x^3\}$ be the standard basis of P_3.

ii. We can view q_1, q_2, q_3 and q_4 by their coordinate vectors in $\mathcal{U}$,

$$\mathbf{u}_1 = [q_1]_{\mathcal{U}} = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_2 = [q_2]_{\mathcal{U}} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_3 = [q_3]_{\mathcal{U}} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_4 = [q_4]_{\mathcal{U}} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

iii. Let us find a basis of $\mathrm{C}(A) = \mathrm{Span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4)$, where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

To do so, transform $\mathbf{A}$ to RREF to obtain

$$\mathbf{A}' = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The columns in $\mathbf{A}$ corresponding to the pivot columns are $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$. So $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ is a basis of $\mathcal{W}$, and $\dim \mathcal{W} = 3$. Thereby answering question (c) too.

(c) $\dim \mathcal{W} = 3$.

Question 6. Let $\mathbf{V}$ be a vector space and let $\{\mathbf{u}_1, \cdots, \mathbf{u}_k\}$ be a linearly independent set. Let $\mathbf{x} \in \mathbf{V}$. Show that, if $\mathcal{U} = \{\mathbf{u}_1, \cdots, \mathbf{u}_k, \mathbf{v}\}$ spans $\mathcal{V}$, then either $\mathbf{v} \in \text{Span}(\{\mathbf{u}_1, \cdots, \mathbf{u}_k\})$ or $\dim \mathcal{V} = k + 1$.

Solution

Suppose that $\text{Span}(\mathcal{U}) = \mathcal{V}$. If $\mathcal{U}$ is linearly independent, then $\mathcal{U}$ is a basis of $\mathcal{V}$, and $\dim \mathcal{V} = k + 1$.

If otherwise, $\mathcal{U}$ is linearly dependent, then $\exists c_1, c_2, \cdots, c_k, c_k + 1 \in \mathbb{R}$ such that

$$c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k + c_{k+1}\mathbf{u}_{k+1} = 0.$$

If $c_{k+1} = 0$, we get $c_1 = \cdots = c_k = c_{k+1} = 0$ because $\{\mathbf{u}_1, \cdots, \mathbf{u}_k\}$ are linearly independent, but then we get that $\mathcal{U}$ is linearly independent, which is absurd. Therefore, $c_{k+1} \neq 0$, hence

$$\mathbf{v} = -\frac{c_1}{c_{k+1}}\mathbf{u}_1 - \cdots - -\frac{c_k}{c_{k+1}}\mathbf{u}_k \in \text{Span}(\{\mathbf{u}_1, \cdots, \mathbf{u}_k\}).$$

Question 7. Consider $\mathcal{U} = \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} m \\ -1 \\ 2 \end{bmatrix} \right\}$. For what values of $m \in \mathbb{R}$ does $\mathcal{U}$ form a basis of $\mathbb{R}^3$? Justify your answer.

Solution

i. Since $\dim \mathbb{R}^3 = 3$, $\mathcal{U}$ is a basis of $\mathbb{R}^3$ if and only if $\mathcal{U}$ is linearly independent.

ii. $\mathcal{U}$ is linearly independent if and only if

$$\mathbf{A} = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3] = \begin{bmatrix} 1 & 1 & m \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

is invertible.

iii. Transform $\mathbf{A}$ to RREF, we obtain $\mathbf{A} \to \mathbf{A}'$ where

$$\mathbf{A}' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

if $m \neq 0$ and

$$\mathbf{A}' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

if $m = 0$.

So $\mathcal{A}$ is invertible if $m \neq 0$, and not invertible for $m = 0$.

Detailed steps:

$$\begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \qquad \begin{bmatrix} 1 & 1 & m \\ 0 & 1 & -1-m \\ 0 & -2 & 2-m \end{bmatrix}$$

$$R_3 \to R_3 + 2R_2 \qquad \begin{bmatrix} 1 & 1 & m \\ 0 & 1 & -1-m \\ 0 & 0 & -3m \end{bmatrix}$$

$$R_1 \to R_1 - R_2 \qquad \begin{bmatrix} 1 & 0 & 1+2m \\ 0 & 1 & -1-m \\ 0 & 0 & -3m \end{bmatrix}$$

If $m = 0$, we stop. If $m \neq 0$, we do $R_3 \to -\frac{1}{3}R_3$, $R_2 \to R_2 + (1+m)R_3$, $R_1 \to R_1 - (1+2m)R_3$ to obtain $\mathbf{I}$.

Question 8. Let $\mathbf{x}_1 = [1, 1, 1]^T$ and $\mathbf{x}_2 = [3, 1, 2]^T$.

(a) Does $\{\mathbf{x}_1, \mathbf{x}_2\}$ span $\mathbb{R}^3$?

(b) Find $\mathbf{x}_3 \in \mathbb{R}^3$ such that $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is basis of $\mathbf{R}^3$.

Solution

(a) No, or otherwise $\dim \mathbb{R}^3 \leq 2$, which is absurd knowing that $\dim \mathbb{R}^3 = 3$.

(b) i. Let $\mathbf{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. We can argue as in Exercise 7 and transform

$$\mathbf{A} = [\mathbf{x}_1\ \mathbf{x}_2\ \mathbf{x}_3] = \begin{bmatrix} 1 & 3 & a \\ 1 & 1 & b \\ 1 & 2 & c \end{bmatrix}$$

to RREF.

ii. Steps to transform $\mathbf{A}$ to RREF

$$\begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \qquad \begin{bmatrix} 1 & 3 & a \\ 0 & -2 & b-a \\ 0 & -1 & c-a \end{bmatrix}$$

$$R_2 \to -\tfrac{1}{2}R_2 \qquad \begin{bmatrix} 1 & 3 & a \\ 0 & 1 & \frac{a-b}{2} \\ 0 & -1 & c-a \end{bmatrix}$$

$$R_3 \to -R_3 + R_2 \qquad \begin{bmatrix} 1 & 3 & a \\ 0 & 1 & \frac{a-b}{2} \\ 0 & 0 & c-\frac{a+b}{2} \end{bmatrix}$$

$$R_1 \to -R_1 - 3R_2 \qquad \begin{bmatrix} 1 & 0 & -\frac{a+b}{2} \\ 0 & 1 & \frac{a-b}{2} \\ 0 & 0 & c-\frac{a+b}{2} \end{bmatrix}$$

If $c - \frac{a+b}{2} \neq 0$, we can continue and transform the matrix to $\mathbf{I}$, showing that $\mathbf{A}$ is invertible.

iii. Choose a, b and c, such that $c - \frac{a+b}{2} \neq 0$, for example $a = 1, b = 1, c = 2$. Then

$$\mathbf{x}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

and $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ is linearly independent, because $\mathbf{A} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3]$ is invertible.

Question 9. Let $\mathcal{W} = \text{Span}\left\{\begin{bmatrix}1\\2\\2\end{bmatrix}, \begin{bmatrix}2\\5\\4\end{bmatrix}, \begin{bmatrix}-1\\-3\\-2\end{bmatrix}, \begin{bmatrix}2\\7\\4\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$. Find a basis of $\mathcal{W}$ and deduce that $\mathcal{W} = \mathbb{R}^3$.

Solution

$\mathcal{W}$ is the column space of the matrix

$$\mathbf{A} = \begin{bmatrix}1 & 2 & -1 & 2 & 1\\ 2 & 5 & -3 & 7 & 1\\ 2 & 4 & -2 & 4 & 0\end{bmatrix}$$

Transforming $\mathbf{A}$ to RREF we get

$$\mathbf{A}' = \begin{bmatrix}1 & 0 & 1 & -4 & 0\\ 0 & 1 & -1 & 3 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

So $\left\{\mathbf{u}_1 = \begin{bmatrix}1\\2\\2\end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix}2\\5\\4\end{bmatrix}, \mathbf{u}_5 = \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$ is a basis of $\mathcal{W}$, and $\dim \mathcal{W} = 3$.

Question 10. (a) Show that $\mathcal{B} = \left\{\mathbf{b}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix}3\\-4\\0\end{bmatrix}, \mathbf{b}_3 = \begin{bmatrix}3\\-6\\3\end{bmatrix}\right\}$ is a basis of $\mathbb{R}^3$.

(b) Suppose that coordinate of $\mathbf{y} = \begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$ in $\mathcal{B}$ are given as $\begin{bmatrix}1\\2\\3\end{bmatrix}$. Calculate the components y_1, y_2 and y_3 of $\mathbf{y}$.

Solution

(a) Transform $\mathbf{A} = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3]$ to RREF and get $\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$, so $\mathbf{A}$ is invertible, and $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ is linearly independent.

(b)

$$[\mathbf{y}]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

and

$$[\mathbf{y}]_{\mathcal{U}} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

where $\mathcal{U}$ is the standard basis of $\mathbb{R}^3$. The standard basis is

$$\left\{ \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

We have the relation

$$[\mathbf{y}]_{\mathcal{U}} = \mathbf{A}[\mathbf{y}]_{\mathcal{B}},$$

where $\mathbf{A} = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$ is the transform matrix from $\mathcal{B}$ to $\mathcal{U}$, so

$$\mathbf{u}_1 = [\mathbf{b}_1]_{\mathcal{U}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_2 = [\mathbf{b}_2]_{\mathcal{U}} = \begin{bmatrix} 3 \\ -4 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_3 = [\mathbf{b}_3]_{\mathcal{U}} = \begin{bmatrix} 3 \\ -6 \\ 0 \end{bmatrix}$$

$$\mathbf{A} = [\mathbf{u}_1\ \mathbf{u}_2\ \mathbf{u}_3] = \begin{bmatrix} 1 & 3 & 3 \\ 0 & -4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

and

$$\mathbf{A}[\mathbf{y}]_{\mathcal{B}} = \begin{bmatrix} 1 & 3 & 3 \\ 0 & -4 & -6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 16 \\ -26 \\ 9 \end{bmatrix}.$$

Question 11. Let X be a vector space with dimension k, and consider two subspace U and V of X. Let $W = U + V = \{z = u + v | u \in U \text{ and } v \in V\}$, $Y = U \cap V$.

(a) Show that Y is a subspace of X.

(b) Show that W is a subspace of X and $\dim W \leq \dim U + \dim V$.

(c) Show that, if $Y = \{0\}$, then $\dim W = \dim U + \dim V$.

Solution

(a) i. $0 \in U$ and $0 \in V$, so $0 \in U \cap V$

ii. If $x \in U \cap V$ and $y \in U \cap V$, then,

$$\begin{cases} x \in U \\ y \in U \end{cases} \quad \text{and} \quad \begin{cases} x \in V \\ y \in V \end{cases}$$

Knowing that U and V are subspace, we have $x + y \in U$ and $x + y \in V$; hence $x + y \in U \cap V$.

iii. Let $k \in \mathbb{R}$ and $x \in U \cap V$. Then $x \in U$ and $x \in V$. Knowing that U and V are subspace, and $kx \in V$, hence $kx \in U \cap V$.

(b) i. W can be shown to be a subspace of X in assignment 3. (Do it again to make the proof.)

ii. Let $\mathcal{B} = \{u_1, \cdots, u_n\}$ be a basis of U. Let $\mathcal{B}' = \{v_1, \cdots, v_m\}$ be a basis of V.

iii. Let $F = \{u_1, \cdots, u_n, v_1, \cdots, v_m\}$. We have $W = \text{Span}(F)$. In fact, let $x \in W = U + V$; there are $u \in U$, $v \in V$, $x = u + v$, and

$$u = k_1 u_1 + \cdots + k_n u_n, k_1, \cdots, k_n \in \mathbb{R},$$

$$v = c_1 v_1 + \cdots + k_n v_m, c_1, \cdots, c_m \in \mathbb{R}.$$

So

$$u + v = k_1 u_1 + \cdots + k_n u_n + c_1 v_1 + \cdots + k_n v_m \in \text{Span}(F).$$

Conversely, if $x \in \text{Span}(F)$, then

$$x = \underbrace{k_1 u_1 + \cdots + k_n u_n}_{\in U} + \underbrace{c_1 v_1 + \cdots + k_n v_m}_{\in V}.$$

So $x \in U + V = W$.

iv. If F is linearly dependent, then we can extract from F a basis of $\text{Span}(F) = W$, and so $\dim W < n + m = \dim U + \dim V$.

(c) i. Suppose that $Y = \{0\}$, that is, if $x \in U \cap V$, then $x = 0$.

ii. Recall that $F = \{u_1, \cdots, u_n, v_1, \cdots, v_m\}$ spans $W = U + V$.

iii. Let $k_1, \cdots, k_n, c_1, \cdots, c_m \in \mathbb{R}$ such that

$$\underbrace{k_1u_1 + \cdots + k_nu_n}_{=u \in U} + \underbrace{c_1v_1 + \cdots + k_nv_m}_{=v \in V} = 0.$$

So $\underbrace{u}_{\in U} = -\underbrace{v}_{\in V}$.

Thus $u \in U \cap V$, hence $u = 0$, $k_1u_1 + \cdots + k_nu_n = 0$, $k_1 = \cdots = k_n = 0$ because $\{u_1, \cdots, u_n\}$ is a basis of U.

Now $c_1v_1 + \cdots + k_nv_m = 0$, and so $c_1 = \cdots = c_m = 0$.

We proved that F is linearly independent,

iv. Knowing that F spanning W and is linearly independent, we get that F is a basis of W. Thus,

$$\dim W = n + m = \dim U + \dim V.$$